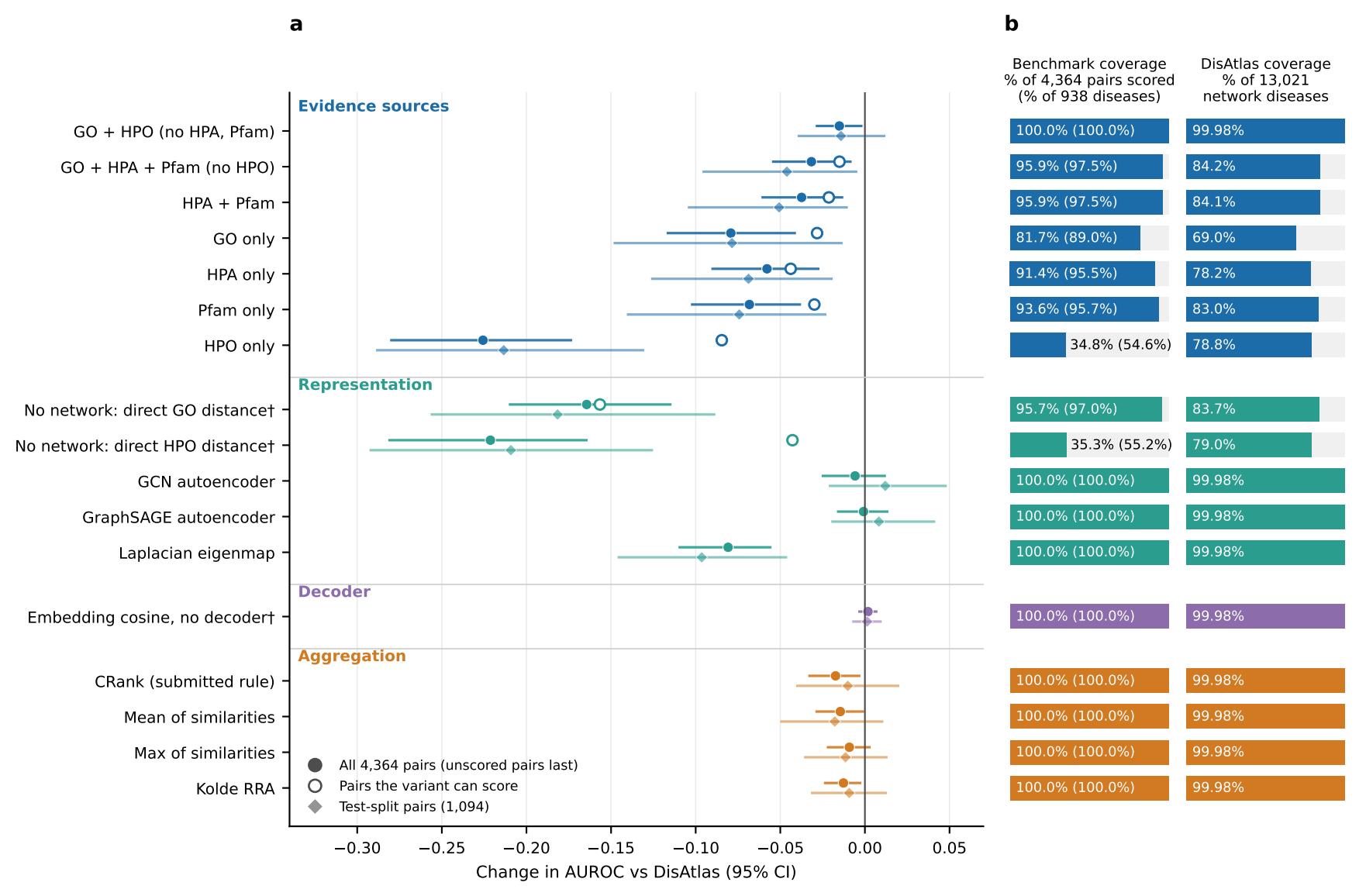
a
b
Evidence sources
GO + HPO (no HPA, Pfam)
GO + HPA + Pfam (no HPO)
HPA + Pfam
GO only
HPA only
Pfam only
HPO only
Representation
No network: direct GO distance†
No network: direct HPO distance†
GCN autoencoder
GraphSAGE autoencoder
Laplacian eigenmap
Decoder
Embedding cosine, no decoder†
Aggregation
CRank (submitted rule)
Mean of similarities
Max of similarities
Kolde RRA
All 4,364 pairs (unscored pairs last)
Pairs the variant can score
Test-split pairs (1,094)
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
0.05
Change in AUROC vs DisAtlas (95% CI)
Benchmark coverage
% of 4,364 pairs scored
(% of 938 diseases)
DisAtlas coverage
% of 13,021
network diseases
100.0% (100.0%)
99.98%
95.9% (97.5%)
84.2%
95.9% (97.5%)
84.1%
81.7% (89.0%)
69.0%
91.4% (95.5%)
78.2%
93.6% (95.7%)
83.0%
34.8% (54.6%)
78.8%
95.7% (97.0%)
83.7%
35.3% (55.2%)
79.0%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%
100.0% (100.0%)
99.98%